THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An AP with $a = -2$ and $d = 0$ is: 1

(a) $0, -2, -4, \ldots$ **(b)** $-2, 0, 2, \ldots$ **(c)** $-2, -2, -2, \ldots$ **(d)** $-2, -4, -6, \ldots$

2. The 15th term of the AP $3, 7, 11, \ldots$ is: 1

(a) 63 **(b)** 59 **(c)** 55 **(d)** 60

3. The sum of the first 100 natural numbers is: 1

(a) 5050 **(b)** 5000 **(c)** 4950 **(d)** 10100

4. A student states: "$1, 4, 9, 16, \ldots$ is an AP because the numbers increase." The correct response is: 1

(a) Incorrect — the differences 3, 5, 7 are not equal

(b) Correct with $d = 3$

(c) Correct

(d) Cannot be determined

5. If $a_n = 5 - 2n$, Student A says $d = -2$; Student B says $d = 5$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The distance between $(a, 0)$ and $(0, a)$ is: 1

(a) a **(b)** $a\sqrt{2}$ **(c)** $\sqrt{a}$ **(d)** $2a$

7. The mid-point of the join of $(0, 0)$ and $(8, -6)$ is: 1

(a) $(4, 3)$ **(b)** $(4, -3)$ **(c)** $(-4, 3)$ **(d)** $(8, -6)$

8. Three vertices of a rectangle are $(0, 0)$, $(4, 0)$ and $(4, 3)$. The fourth vertex is: 1

(a) $(4, 4)$ **(b)** $(0, 3)$ **(c)** $(3, 0)$ **(d)** $(3, 4)$

9. A student states: "The origin is equidistant from $(3, 4)$ and $(-4, 3)$." The correct response is: 1

(a) Correct — both distances are 5 **(b)** Correct only for the first point

(c) Incorrect **(d)** Cannot be determined

10. For the point dividing $(2, 4)$ and $(6, 8)$ in the ratio 1 : 1, Student A gets $(4, 6)$; Student B gets $(3, 5)$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Is 0 a term of the AP $31, 28, 25, \ldots$? Justify. 2

12. Find the sum of the first 25 terms of an AP whose nth term is $7 - 3n$. 2

13. Find the point on the x-axis equidistant from $(7, 6)$ and $(3, 4)$. 2

14. Find the coordinates of P on AB such that $\frac{AP}{PB} = \frac{2}{3}$, where A(−1, 3) and B(9, 8). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How many terms of the AP $54, 51, 48, \ldots$ must be taken so that their sum is 513? Explain the double answer. 3

16. Find the centre of the circle passing through $(6, -6)$, $(3, -7)$ and $(3, 3)$. 3

17. Find four numbers in AP whose sum is 20 and the sum of whose squares is 120. 3

OR

If $a_n = 9 - 5n$, show that the sequence is an AP and find S_{15}. 3

18. Find a relation between x and y such that (x, y) is equidistant from A(5, 1) and B(−1, 5). 3

OR

Find the lengths of the medians of the triangle with vertices A(0, −1), B(2, 1) and C(0, 3). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** An auditorium has 20 seats in the first row and each next row has 2 more seats than the previous one. There are 30 rows. 5

(i) Find the number of seats in the 30th row.

(ii) Find the total number of seats.

(iii) Which row has 50 seats?

OR

If the sum of the first p terms of an AP equals the sum of its first q terms ($p \neq q$), prove that the sum of its first $(p + q)$ terms is zero. Verify for the AP $24, 21, 18, \ldots$ with $p = 4,\ q = 13$. 5

20. **Case study:** A park ABCD has corners A(1, 1), B(7, 1), C(7, 9) and D(1, 9). 5

(i) Find the diagonal AC.

(ii) A fountain is at the centre of the park. Find its position.

(iii) Two lamp posts trisect side AB. Find their positions.

(iv) Verify that the fountain is equidistant from A and C.

OR

A(4, 3) and B(x, 5) lie on a circle with centre O(2, 3). 5

(i) Find x.

(ii) Find the other end of the diameter through A.

(iii) Check whether $(2, 1)$ lies on the circle.

END OF QUESTION PAPER — Total: 40 Marks